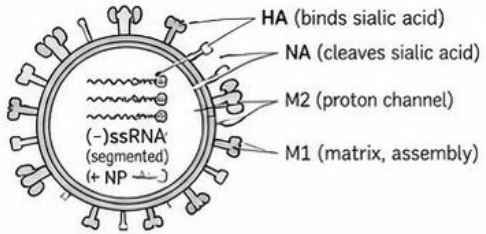
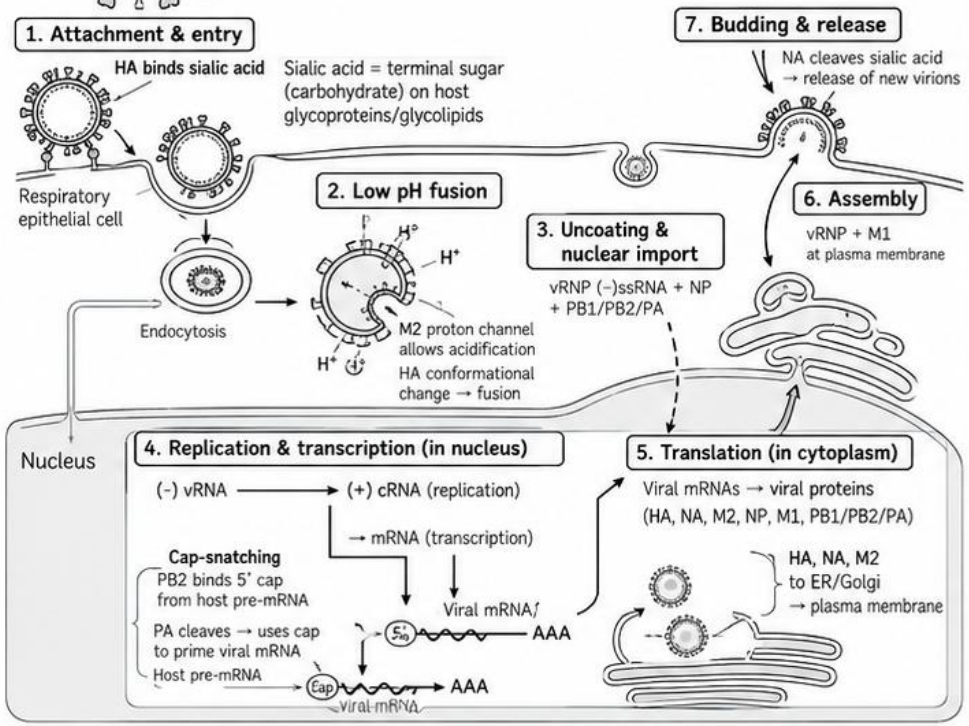
# Influenza A virus (Orthomyxoviridae)

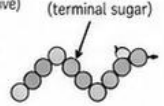



**Key features**
- Enveloped, (-)ssRNA (8 segments)
- HA, NA, M2, M1, NP, PB1/PB2/PA
- Replicates in nucleus
- Infects respiratory epithelium
- Human: α2,6 sialic acid (upper RT)
- Avian: α2,3 sialic acid (lower RT)

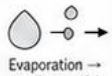



| Viral protein | Location | Function | Mutation/inhibitor → expected result (examples) |
|---|---|---|---|
| **HA** | Envelope | Binds sialic acid, mediates fusion (low pH) | HA mutation (no SA binding) → no entry/infection<br>Acidification block → no fusion |
| **NA** | Envelope | Cleaves sialic acid, release of virions | NA inhibitor (oseltamivir) → virions remain attached, reduced spread |
| **M2** | Envelope | Proton channel, acidifies virion | M2 blocker (amantadine) → no uncoating |
| **M1** | Matrix (inner) | Virion assembly, nuclear export | M1 mutation → defective assembly/release |
| **NP** | Nucleocapsid | Binds vRNA | NP mutation → unstable vRNP, ↓ replication |
| **PB1** | Polymerase | RNA-dependent RNA polymerase | Polymerase inhibitor → no replication/transcription |
| **PB2** | Polymerase | Binds 5' cap (cap-snatching) | PB2 mutation → no viral mRNA synthesis |
| **PA** | Polymerase | Endonuclease (cleaves host pre-mRNA) | PA mutation → no cap-snatching |

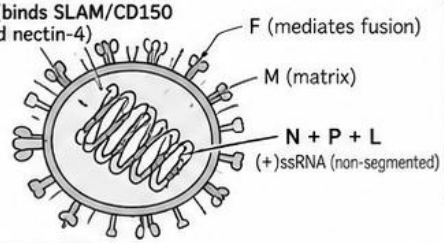
**Sialic acid & carbohydrates**
- Sialic acid is a carbohydrate (sugar derivative) on host cell surface.
- Enzyme that cleaves carbohydrates (e.g., neuraminidase) → removes sialic acid → prevents HA binding and infection.



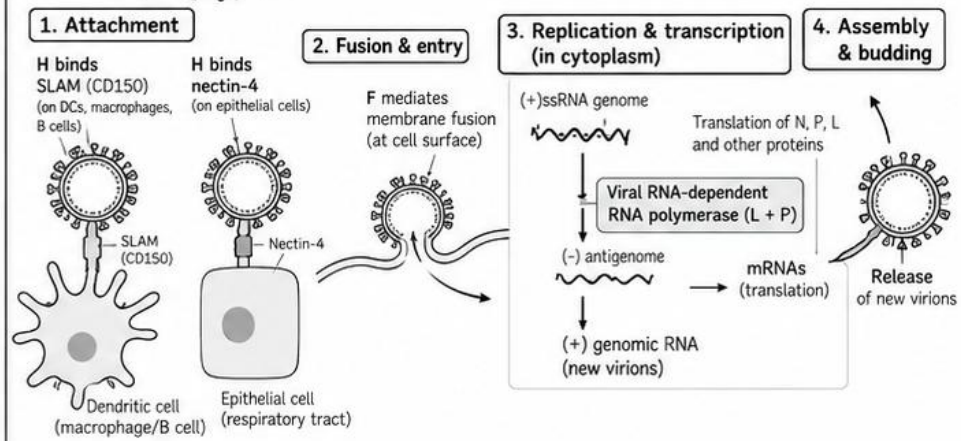

**Transmission & tropism**
- Aerosolized particles < droplets → remain in air longer, reach lower respiratory tract (alveoli).
- Lower RT infection → closer to blood capillaries → higher chance of systemic spread.
- Human: α2,6 SA (upper RT), Avian: α2,3 SA (lower RT).
- Cold/dry conditions → droplets evaporate → smaller, remain airborne longer → ↑ transmission.




# Measles virus (Paramyxoviridae)

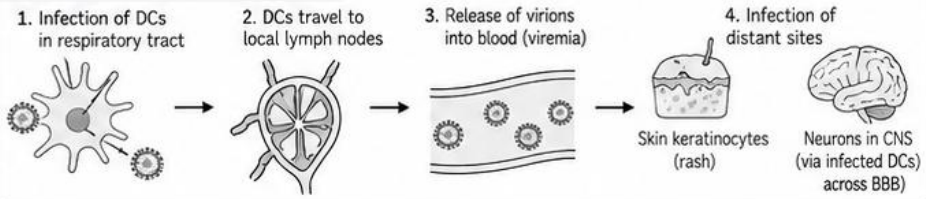



**Key features**
- Enveloped, (+)ssRNA (non-segmented)
- H (attachment), F (fusion), N, P, L, M
- Replicates in cytoplasm
- Initial infection of immune cells (DCs, macrophages)
- Systemic spread via lymph nodes and blood
- Rash (skin) and CNS infection can occur




| Viral protein | Location | Function | Mutation → expected result (examples) |
|---|---|---|---|
| **H** | Envelope | Binds SLAM (CD150) on immune cells; nectin-4 on epithelial cells | • H mutation (no SLAM binding) → no DC infection, no viremia, no rash, no respiratory shedding.<br>• H mutation (still binds SLAM, no nectin-4 binding) → can enter blood, milder/normal rash. |
| **F** | Envelope | Mediates membrane fusion | • F mutation → no entry |
| **N** | Nucleocapsid | Binds genomic RNA | • N mutation → ↓ replication |
| **P** | Polymerase cofactor | Cofactor for L | • P mutation → ↓ replication |
| **L** | Polymerase | RNA-dependent RNA polymerase | • L mutation → no replication |
| **M** | Matrix | Virion assembly and budding | • M mutation → defective assembly/release |

**Pathogenesis & spread (summary)**

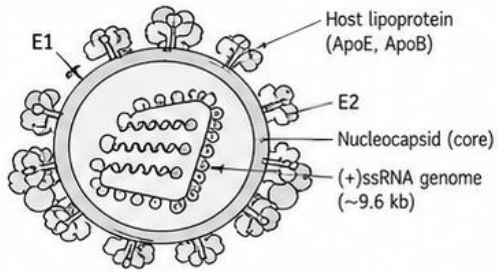



**SLAM, nectin-4 and mutant phenotypes**

| | |
|---|---|
| H cannot bind SLAM (mutant #1) | • No infection of DCs/macrophages<br>• No travel to lymph nodes → no viremia<br>• No rash (skin) and no respiratory shedding |
| H cannot bind nectin-4 (mutant #2) | • Can still bind SLAM → viremia occurs<br>• Mild rash (less skin infection)<br>• No respiratory shedding (cannot infect respiratory epithelium) |

**Clinical features**
- Respiratory symptoms (from epithelial infection)
- Swollen lymph nodes (immune activation)
- Rash (skin infection)
- Can cause CNS infection (via infected DCs)
- Highly contagious (measured by $R_0$)

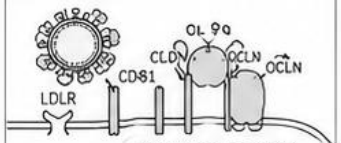
# Hepatitis C virus (*Flaviviridae*)




**Key features**

- Enveloped, (+)ssRNA (non-segmented)
- E1/E2 (attachment & fusion)
- Replicates in cytoplasm (membranous web)
- Associates with host lipids/lipoproteins
- High mutation rate (error-prone RdRP)
- Establishes persistent infection
- Transmitted via blood (also cell–cell)
- Can cause chronic hepatitis, cirrhosis, HCC

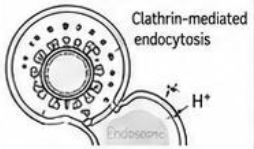
## 1. Attachment




Hepatocyte

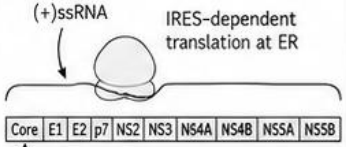
E2 binds SR-B1, CD81
Tight junction proteins CLDN1 and OCLN required for entry

## 2. Entry & uncoating



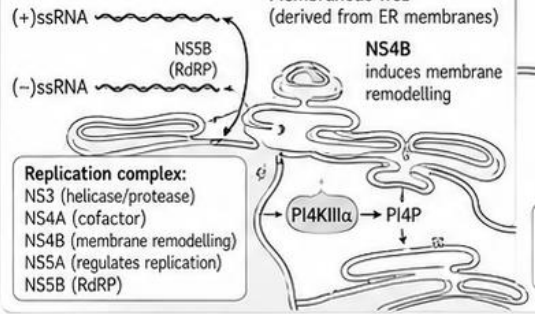

Low pH → conformational change in E1/E2 → fusion
Uncoating → release of (+)ssRNA into cytoplasm

## 3. Translation



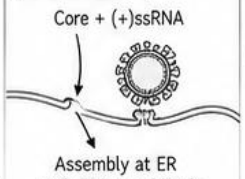

Polyprotein
Host & viral proteases (NS2, NS3/4A) → cleavage

## 4. Replication (membranous web)



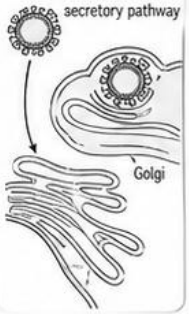

**Replication complex:**
NS3 (helicase/protease)
NS4A (cofactor)
NS4B (membrane remodelling)
NS5A (regulates replication)
NS5B (RdRP)

## 5. Assembly




Hijacks host PI4KIIIα → ↑ PI4P → recruits replication machinery

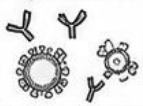
## 6. Release




| Viral protein | Location | Function | Mutation/inhibitor → expected result (examples) |
|---|---|---|---|
| **Core** | Nucleocapsid | Binds RNA, assembly, modulates host signals | Core mutation → ↓ assembly; affects pathogenesis |
| **E1** | Envelope | Heterodimer with E2, fusion | E1 mutation → defective entry/fusion |
| **E2** | Envelope | Receptor binding (SR-B1, CD81) | E2 mutation → no entry; antibody neutralization |
| **p7** | Viroporin | Ion channel, virion assembly | p7 inhibitor → ↓ assembly/release |
| **NS2** | NS protease | Autoprotease, assembly | NS2 mutation → polyprotein not cleaved |
| **NS3** | Protease/helicase | Cleaves polyprotein, unwinds RNA | NS3 protease inhibitor → no replication |
| **NS4A** | Cofactor | Cofactor for NS3 | NS4A mutation → ↓ NS3 activity |
| **NS4B** | Membrane mod. | Induces membranous web | NS4B mutation → defective replication complex |
| **NS5A** | Phosphoprotein | Regulates replication, assembly | NS5A inhibitor → ↓ replication; ↓ virion production |
| **NS5B** | RdRP | RNA-dependent RNA polymerase | NS5B inhibitor (nucleoside/NSSB) → no RNA synthesis |

## HCV immune evasion (antibody response)

**1. High mutation rate (error-prone RdRP)**



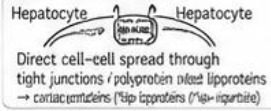
Rapid generation of escape mutants → antibodies become less effective

**2. Association with host lipids/lipoproteins**



New virions circulate with host lipoproteins (ApoE, ApoB) → mask viral epitopes → reduced antibody binding

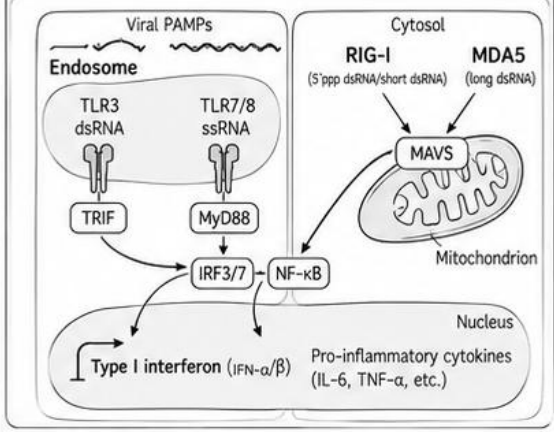
**3. Cell–cell transmission**



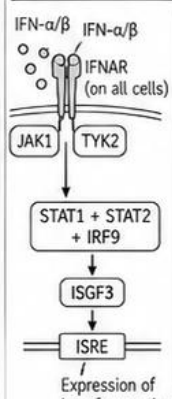

Direct cell–cell spread through tight junctions / polyprotein-linked lipoproteins → cellular receptors (host lipoproteins / viral glycoproteins)

# Host antiviral responses & key concepts

## 1. Innate immune recognition




## 2. Type I interferon responses (IFN-α/β)



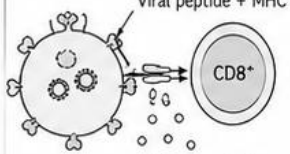

Expression of interferon-stimulated genes (ISGs)

**Key ISGs and functions**

- **PKR** — Inhibits translation (activated by dsRNA)
- **OAS → RNase L** — Degrades viral and cellular RNA
- **Mx proteins** — Inhibit viral transcription/replication
- **IFIT proteins** — Bind viral RNA, inhibit translation
- **Tetherin** — Inhibits virion release

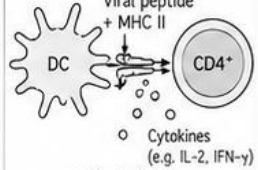
## 3. Effector immune responses (adaptive immunity)

**Cytotoxic T lymphocytes (CTLs)**



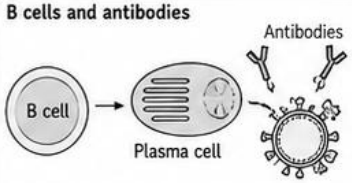

Release perforin and granzymes → kill infected cell

**Helper T cells (CD4+)**




- Activate CTLs
- Activate B cells
- Activate macrophages

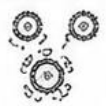
**B cells and antibodies**




- Neutralize virus (block attachment/entry)
- Opsonization (enhances phagocytosis)
- Activate complement
- Antibody-dependent cellular cytotoxicity (ADCC)

## 4. Viral evasion of host immunity (general strategies)

**Rapid mutation (antigenic drift)**



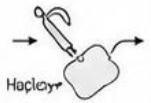
Escape from neutralizing antibodies and T cell responses

**Antigenic shift (reassortment, e.g. IAV)**



- New HA/NA → little/no pre-existing immunity

**Inhibit innate signaling (e.g. HCV NS3/4A)**



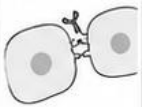
Blocks MAVS/TRIF, inhibits IFN production

**Hide from antibodies (e.g. HCV lipoprotein)**



Mask viral epitopes

**Cell–cell spread**



Avoids extracellular antibodies

**Latency/persistent infection**



Low antigen expression → reduced immune detection

## Key concepts for experimental/prediction questions

| | |
|---|---|
| **Carbohydrates & sialic acid** | • Sialic acid = terminal sugar on host glycoproteins/glycolipids (a carbohydrate derivative)<br>• Removing carbohydrates (e.g. neuraminidase) → removes sialic acid → prevents IAV attachment |
| **Viral receptors** | • IAV: sialic acid (α2,6 human upper RT; α2,3 avian lower RT) • MeV: SLAM (CD150), nectin-4<br>• HCV: SR-B1, CD81, CLDN1, OCLN, LDLR (E2) |
| **Nuclear vs cytoplasmic replication** | • DNA viruses without their own DNA polymerase (e.g. small ssDNA) → use host nuclear machinery<br>• RNA viruses (except retroviruses) → encode RdRP and replicate in cytoplasm |
| **Viral shedding** | • Respiratory shedding: virus released from respiratory tract<br>• Fecal shedding: virus released in feces (if GI infection)<br>• Shedding site depends on sites of infection, not just entry site |
| **Transmission & environment** | • Aerosol (small, stays airborne longer, can reach lower RT) vs droplet (larger, short range)<br>• Dry/cold conditions → droplets evaporate → smaller particles → increased airborne transmission |
| **Systemic spread** | • Often via infection of immune cells (e.g. MeV infection of DCs → lymph nodes → viremia)<br>• Swollen lymph nodes ≠ necessarily immune cell infection (can be immune activation) |